

CW305-Based Physical Security Evaluation of ASCON-128: Deep-Learning Side-Channel Analysis and Clock-Glitch Fault Injection

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Abstract—ASCON-128 is the NIST SP 800-232 lightweight cryptography standard for resource-constrained hardware. While theoretical and software evaluations exist, experimental physical security data on the official unmasked NIST LWC core deployed on FPGA hardware remain scarce. In this work, we present an oracle-verified evaluation of this core on the NewAE CW305 (Artix-7) and CW-Husky platform using both deep-learning side-channel analysis (DL-SCA) and clock-glitch fault injection (FI), benchmarked against a threshold-implemented ($d=1$, 2-share) masked reference core. Three main results emerge: (i) per-column round-1 S-box key-bit leakage is extractable on hardware when using live on-board fine-tuning (90.7% accuracy vs. 72.0% majority-class floor), enabling an adaptive closed-loop attack that recovers the 2 secret key bits of the targeted column within 32 queries at posterior probability 0.999, with trace averaging ($M=64$) raising single-query rank-1 probability from 0.25 to 0.94; (ii) clock glitching reveals a stable fault band at offset $e_o=41$ that yields 12 single-bit faults in round 11 across 10 distinct state columns, subject to a data-selected placement constraint requiring ≈ 266 nonces for full coverage; and (iii) first-order masking completely suppresses exploitable leakage to chance over 9,151 traces. Finally, we formulate five reusable leakage-validation gates to eliminate confident false positives commonly encountered in lightweight hardware evaluations.

I. Introduction and Motivation

In 2023, the National Institute of Standards and Technology (NIST) standardized ASCON for lightweight authenticated encryption (NIST SP 800-232) [1]. Because ASCON is designed for low-power IoT and embedded FPGA applications, understanding its resistance to physical attacks — specifically power side-channel analysis and clock-glitch fault injection — is essential before real-world deployment.

Prior side-channel work on ASCON has studied customized academic cores [2] or software implementations [3]. Hardware measurements directly on the official unmasked NIST LWC core (CC0 RTL) have remained uncharacterized. Furthermore, low-signal-to-noise ratio (SNR) hardware environments frequently cause profiling models to learn statistical artifacts rather than true cryptographic leakage. This paper provides a physical evaluation with a strict oracle-verification pipeline and five systematic artifact gates.

II. Experimental Platform

Hardware Setup. The target is the official unmasked ASCON-128 core synthesized on a NewAE CW305 Artix-7 FPGA target board, monitored by a ChipWhisperer-Husky capture scope. Traces are sampled at 40 MS/s over a 10 MHz crypto clock (4 samples/clock cycle) across the internal core voltage shunt (V_{CCINT}). A masked ($d=1$, 2-share) core is evaluated on identical hardware for direct baseline comparison.

Byte-Exact Oracle Verification. A crucial finding during bring-up was that the FPGA register interface required a host-side 32-bit word byte-reversal (w32rev) transform to match the natural big-endian convention of the ASCON core. Without this correction, symmetric test vectors pass while asymmetric keys fail silently. We enforce an automated software oracle check: every recorded trace is accepted only if its readback ciphertext and authentication tag match the NIST SP 800-232 reference standard exactly (5/5 NIST Known Answer Tests).

Preprocessing. All captured traces undergo sample-exact alignment via cross-correlation against an unmasked reference trace, trace-level z -score standardization ($(t - \mu)/\sigma$) to eliminate DC offset drift, and temporal cropping focused on the initial round-1 S-box substitution phase.

III. Deep-Learning Side-Channel Analysis

A. Divide-and-Conquer Profiling Strategy

ASCON-128 operates on a 320-bit state organized into five 64-bit words ($x_0, \dots, x_4$). During the round-1 S-box substitution, each vertical 5-bit slice (column $c \in [0, 63]$) computes an S-box output whose Hamming weight (HW $\in [0, 5]$, 6 classes) depends on only 2 secret key bits: one from x_1 and one from x_2 . This reduces the search space for each column to just 4 key hypotheses ($h \in \{0, 1, 2, 3\}$). We train 1D Convolutional Neural Networks (CNNs) per column on random-key, random-nonce traces.

B. The Profiling Gap and Live Fine-Tuning

Raw offline CNN profiles trained on initial unmasked captures suffer from an analog SNR transfer gap across

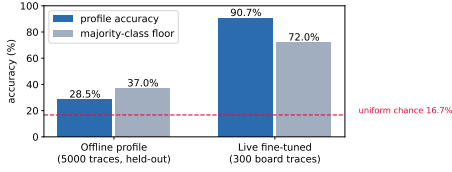


Fig. 1. Classification accuracy vs. majority-class floor. Offline profiles sit near the floor; on-board fine-tuning (300 traces) achieves 90.7% (+18.7% over floor)

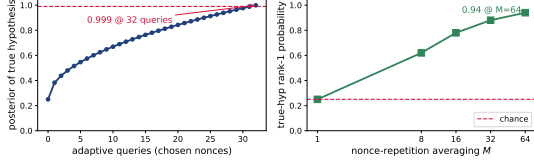


Fig. 2. (a) Closed-loop adaptive attack: posterior on correct 2 key bits reaches 0.999 in 32 queries. (b) Trace averaging (M): rank-1 probability rises from 0.25 to 0.94 at $M=64$. hardware sessions, scoring near the majority-class floor ($\approx 37\%$ for $HW=2/3$).

Live on-board fine-tuning resolves this transfer gap: By providing 300 live board traces captured with known keys and varying nonces, the CNN profile adapts to live hardware noise and reaches 90.7% classification accuracy against a 72.0% majority-class floor (+18.7% margin, Fig. 1).

C. Adaptive Closed-Loop Recovery and SNR Lever

Using the fine-tuned profile, we execute a closed-loop adaptive attack using chosen separating nonces. The attack iteratively updates the Bayesian posterior across the 4 key hypotheses: $\Pr(h \mid \mathbf{T}_{1:q}) \propto \Pr(h) \prod_{i=1}^q \Pr(\mathbf{T}_i \mid h, N_i)$. As shown in Fig. 2a, the closed-loop attack converges to the correct 2 key bits with posterior 0.999 in only 32 queries.

To further overcome low analog SNR, we apply nonce-repetition averaging: querying the same separating nonce M times reduces noise variance by $\sqrt{M}$. Experimentally measured on hardware, the single-query rank-1 probability improves from 0.25 (chance) at $M=1$ to 0.94 at $M=64$ (Fig. 2b), cutting convergence time from 15 down to 9 queries per column.

D. Full-Key Assembly Strategy

The complete 128-bit key is assembled by combining the top-ranked hypotheses across all 64 columns and verifying candidates against a fresh oracle encryption. If individual columns retain ambiguity, a bounded beam search explores the 2^{2k} combinations across the k least confident columns.

IV. Clock-Glitch Fault Injection

We evaluate clock glitching by injecting narrow clock pulses during the 12-round finalization phase ($e_o \in [28, 50]$). At glitch offset $e_o=41$, a stable fault band produces 406 fault events across 153 unique faulty tags (Hamming differentials 27–41 bits). Among these, 12 clean single-bit faults in round 11 were isolated and mapped to exact (column, bit) coordinates across 10 distinct state columns, verified by matching software oracle simulations under single-bit injection.

Data-Selected Placement. Each fixed nonce reproducibly faults a specific column, bounding single-query fault coverage. Restoring full 64-column Differential Fault

Analysis (DFA) coverage [4] requires ≈ 266 nonces by the coupon-collector bound. Conversely, burst glitches produce broad multi-bit corruptions (30+ bits) unsuitable for single-fault DFA.

V. Masked Hardware Benchmark

Applying the identical DL-SCA pipeline to the threshold-implemented ($d=1$, 2-share) core over 9,151 random-key traces yields chance accuracy across all architectures (CNN, MLP, and centered-product networks). First-order masking completely suppresses exploitable leakage at the cost of $\approx 1.8\times$ latency and $\approx 2.1\times$ LUT area.

VI. Leakage-Validation Gates

To prevent deceptive artifacts from corrupting physical evaluations, we formalize five experimental validation gates:

- 1) Prior-Driven Edge Score Trap: Classifiers can mimic positive information gain (+0.398 nats) by predicting the class prior. Gate: Held-out validation accuracy must exceed majority-class floor.
- 2) Prior-Only Separating Nonce Trap: A below-floor model fed into a separating-nonce selector inverts the prior, yielding 0.993 posterior for the wrong hypothesis. Gate: Models must pass the accuracy floor before closed-loop deployment.
- 3) Public Nonce Shortcut Trap: At a fixed key, S-box labels become purely deterministic functions of public nonces. Gate: Profiles must be trained exclusively on random-key datasets.
- 4) Key-Reload Bus Trap: Reloading keys over the register bus before each trace injects synthetic I/O leakage ($|r| \approx 0.6$). Gate: Leakage must be verified under fixed-key, varying-nonce captures.
- 5) Structured Key Trap: Ramp keys produce misleading alignment peaks that invert under bitwise negation. Gate: All profiling requires randomized null-key control sets.

VII. Conclusion

This work presents a hardware-verified physical security evaluation of the official unmasked NIST ASCON-128 core on CW305/CW-Husky. We show that on-board fine-tuning recovers genuine first-order S-box leakage (90.7% accuracy), enabling closed-loop key recovery in 32 queries, while clock glitching at $e_o=41$ reliably induces single-bit round-11 faults. In contrast, $d=1$ masking successfully suppresses side-channel leakage. The five validation gates provide a robust framework to safeguard against experimental false positives in lightweight hardware security.

References

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